Engineering Calculations

1. Payload Capacity Calculation

The robotic arm is designed to safely lift lightweight objects in cold storage environments.

Formula:

$$P_{safe} = P_{expected} \times Safety\ Factor$$

- Expected Payload = 100 g
- Safety Factor = 1.5

Calculation:

$$P_{safe} = 100 \times 1.5 = 150\ g$$

Result:

Safe Payload Capacity = 150 g

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2. Torque Calculation for Robotic Arm

Torque required for lifting operation.

Formula:

$$T = F \times r$$

Where:

- Force:

$$F = m \times g$$

Given:

- Mass = 0.2 kg
- Gravity = 9.81 m/s²
- Arm Length = 0.5 m

Force Calculation:

$$F = 0.2 \times 9.81 = 1.962\ N$$

Torque Calculation:

$$T = 1.962 \times 0.5 = 0.981\ N \cdot m$$

Equivalent Servo Torque:

≈ 10 kg/cm

This validates the use of MG995 servo motors.

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3. Power Consumption Calculation

Formula:

$$P = V \times I$$

For one MG995 Servo Motor:

- Voltage = 6V
- Current = 2.5A

Power for one servo:

$$P = 6 \times 2.5 = 15\ W$$

For 3 Servo Motors:

$$P_{total} = 3 \times 15 = 45\ W$$

Result:

Total Servo Power Consumption = 45 W

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4. Motor Power Calculation

Formula:

$$P_{motor} = \frac{T \times \omega}{1000}$$

Given:

- Torque = 0.981 N·m
- Angular Velocity = 2.05 rad/s

Calculation:

$$P_{motor} = \frac{0.981 \times 2.05}{1000} = 2.05\ W$$

Result:

Motor Power = 2.05 W

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5. Battery Energy Calculation

Formula:

$$Energy = Battery\ Capacity \times Voltage$$

Given:

- Battery Capacity = 2.6 Ah
- Voltage = 7.4 V

Calculation:

$$Energy = 2.6 \times 7.4 = 19.24\ Wh$$

Result:

Battery Energy = 19.24 Wh

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6. Battery Backup Time Calculation

Formula:

$$Battery\ Life = \frac{Battery\ Energy}{Power\ Consumption}$$

Calculation:

$$Battery\ Life = \frac{19.24}{45} = 0.43\ hours$$

Convert to minutes:

$$0.43 \times 60 = 26\ minutes$$

Result:

Estimated Battery Backup = 26 Minutes

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7. Stress Analysis

Used for validating robotic arm structural safety.

Formula:

$$\sigma = \frac{F}{A}$$

Given:

- Load Force:

$$F = 1.962\ N$$

Assume:

- Cross-sectional Area of arm = 200 mm²

Convert area:

$$A = 200 \times 10^{-6}\ m^2$$

Calculation:

$$\sigma = \frac{1.962}{200 \times 10^{-6}} = 9810\ Pa$$

Result:

Stress Developed = 9.81 kPa

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8. Speed Calculation of Robot

Formula:

$$Velocity = RPM \times Circumference$$

Assume:

- Wheel Diameter = 70 mm
- Motor Speed = 100 RPM

Wheel Circumference:

$$C = \pi d = 3.14 \times 0.07 = 0.2198\, m$$

Linear Speed:

$$V = 100 \times 0.2198 = 21.98\, m/min$$

Convert:

$$V = \frac{21.98}{60} = 0.366\, m/s$$

Result:

Robot Speed ≈ 0.36 m/s

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9. Efficiency Calculation

From report:

- Battery optimization improved efficiency by 20%

Formula:

$$Efficiency\ Improvement = \frac{Improved - Original}{Original} \times 100$$

Assume:

- Original battery life = 26 min
- Improved = 31.2 min

Calculation:

$$\frac{31.2 - 26}{26} \times 100 = 20\%$$

Result:

Efficiency Improvement = 20%

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10. Wheel RPM Calculation

Formula:

$$RPM = \frac{60V}{\pi d}$$

Given:

- Velocity = 0.366 m/s
- Diameter = 0.07 m

Calculation:

$$RPM = \frac{60 \times 0.366}{3.14 \times 0.07} = 99.8$$

Result:

Wheel RPM ≈ 100 RPM

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11. Center of Gravity Calculation

Formula:

$$x_{cg} = \frac{\Sigma mx}{\Sigma m}$$

Assume:

Component	Mass (kg)	Distance (m)
Battery	0.12	0.1
Arm	0.30	0.25
Chassis	0.50	0.15

Calculation:

$$x_{cg} = \frac{(0.12 \times 0.1) + (0.3 \times 0.25) + (0.5 \times 0.15)}{0.92}$$

Result:

$$x_{cg} = 0.176\ m$$

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12. Load Distribution Calculation

Formula:

$$Load\ per\ wheel = \frac{Total\ Weight}{Number\ of\ Wheels}$$

Assume:

- Total Weight = 1.2 kg
- Wheels = 4

Calculation:

$$\frac{1.2}{4} = 0.3\ kg$$

Result:

Load per Wheel = 0.3 kg